

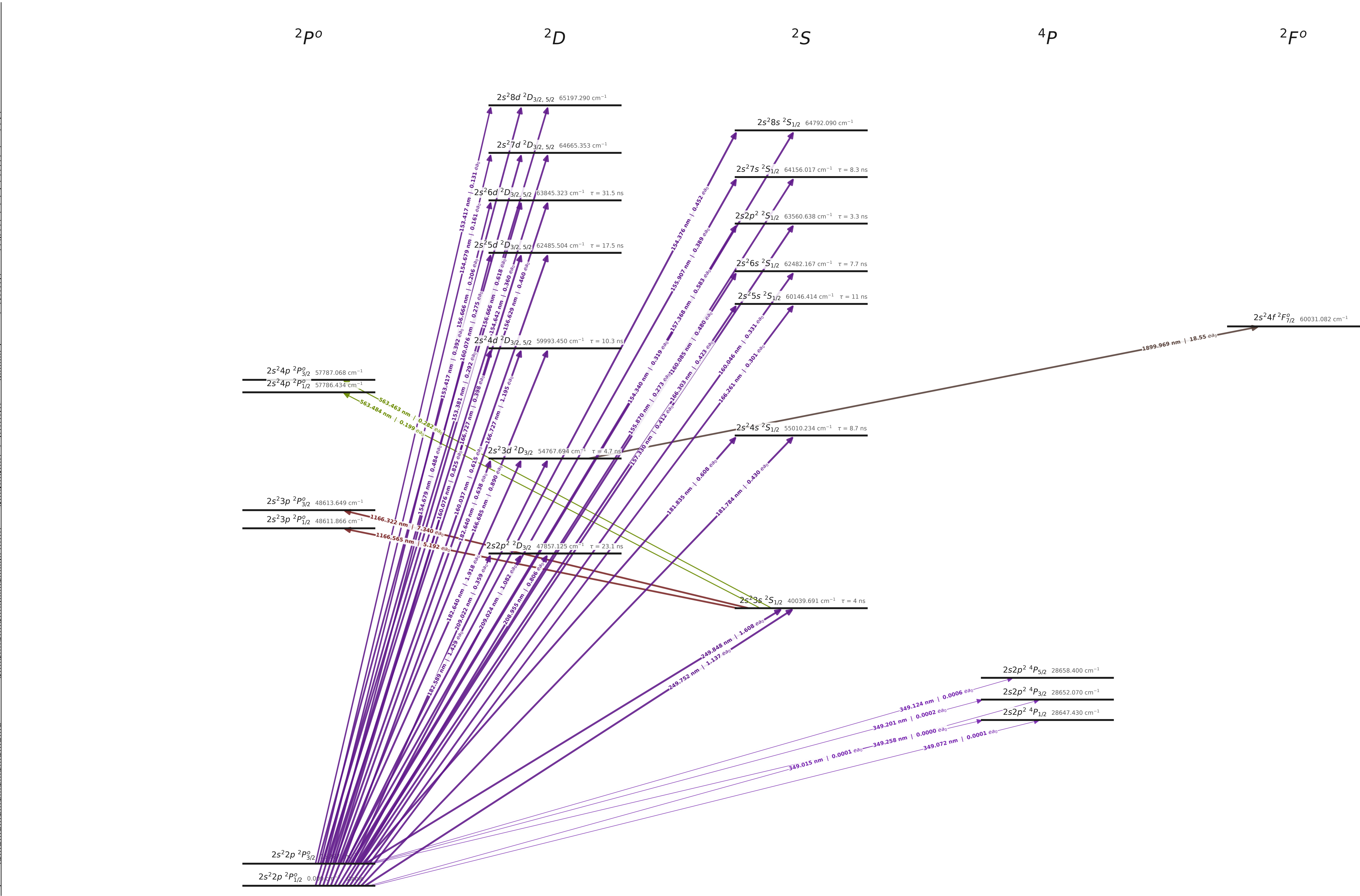
B (B I). Strongest classified transitions below 2 μm (NIST): 31 levels, 45 lines.

B⁺ ionisation limit 66928.04 cm⁻¹ = 8.29802 eV (149.414 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle J \| er \| J' \rangle$ in ea_0
 Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.

Arrow width grows with the Einstein coefficient A . Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.

Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle J \ \mathbf{U} \ J' \rangle $ (ea ₀)	A (s ⁻¹)	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$2s^2 2p \ ^2P^o_{1/2} - 2s^2 3s \ ^2S_{1/2}$	249.7523	249.6770	1200.359034	1.137	8.400e+07	39.79	33.6	249.75 nm
$2s^2 2p \ ^2P^o_{3/2} - 2s^2 3s \ ^2S_{1/2}$	249.8476	249.7723	1199.9014	1.608	1.680e+08	39.79	67.2	249.85 nm

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.

$\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
$2s^2 2p^2 p_{1/2}^o$	366.0765(15)	-	F=1: -457.60 F=2: +274.56
$2s^2 2p^2 p_{3/2}^o$	73.3496(4)	2.6927(10)	F=0: -271.69 F=1: -201.04 F=2: -57.03 F=3: +165.71
$2s^2 3s^2 s_{1/2}$	238.0(4)	-	F=1: -297.50 F=2: +178.50

Measured A, B constants; sources in the data file.

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$2s^22p\ ^2P_{1/2}^o - 2s^23s\ ^2S_{1/2}$	249.752	10-11	5031.3(20)

$$\text{Shift} = \nu(\text{first isotope}) - \nu(\text{second isotope}).$$